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1. Sorting Babies by hair color, Brown -> Purple

(a) Constraints on the Two Pointers

We line the babies up in an array in random order (Brown or Purple).

We use two pointers:

- i = scanning pointer
- j = boundary pointer (next position for a brown baby)

Both pointers only move forward and never backtrack.

b) Thought Process

- In place:
  - We do not use another array.
  - We only swap babies inside the same array.
- Linear Time:
  - Each baby is examined once.
  - Both pointers move forward at most n times.
- One-Pass:
  - Do not need to backtrack through the array of babies

c) Using Hoarse Partition

```
hairSorterBrowntoPurple(arr):

    i = 0
    j = arr.length - 1

    while (true):

        // Move i right until we find Purple
        while (i < arr.length AND arr[i] == Brown):
            i++

        // Move j left until we find Brown
```

```
        while (j >= 0 AND arr[j] == Purple):
            j--

        // If pointers crossed, sorting is done
        if (i >= j):
            break

        // Swap misplaced babies
        swap(arr[i], arr[j])

    return arr
```

2. Sorting Babies by hair color Brown -> Purple -> Black

- Using three pointers

```
hairSorterBrowntoPurpletoBlack(arr):

// low = next position where a Brown baby should go
// mid = current scanning pointer
// high = next position where a Black baby should go

low = 0
mid = 0
high = arr.length - 1

while (mid <= high):

// Case 1: Brown should go to the front section
if (arr[mid] == Brown):
swap(arr[low], arr[mid])
low++
mid++

// Case 2: Purple is already in the correct middle section
else if (arr[mid] == Purple):
mid++

// Case 3: Black should go to the back section
else: // arr[mid] == Black
swap(arr[mid], arr[high])
high--

return arr
```

3 way partition

Dutch National Flag Prob by Dijkstra